3.2 Medical Requirements Overview

TABLE 3.2: MEDICAL REQUIREMENTS OVERVIEW

MEDB# and Title:	MEDB 2.4 Methicillin Resistant Staphylococcus aureus (MRSA) Nasal Screen and Suppression
Sponsor:	Medical Operations
Discipline:	Immunology/Infectious Disease
Category:	Medical Requirements (MR)
References:	Medical Evaluation Documents (MED) Volume B, Section 2.4
Purpose/Objectives:	Personnel will perform microbiological evaluation of crewmembers to complement environmental monitoring of the ISS cabin environment to determine transfer or acquisition of Methicillin-Resistant *Staphylococcus aureus* (MRSA) and assess the presence of this microorganism in the environment.
Measurement Parameters:	Nasal Screen for Methicillin Resistant *Staphylococcus aureus* (MRSA)
Deliverables:	Pre-and Post-flight medical assessment test (MAT) report of results to Flight Surgeon
Flight Duration:	$\geq$30 days
Number of Flights:	All
Number and Type of Crew Members Required:	All primary ISS crewmembers. Back-up crew will only complete preflight MATs greater than L-30 days unless specifically waived by crew surgeon. If crew swap does occur, back-up crew will complete all preflight and post-flight MATs.
Other Flight Characteristics:	N/A

3.3 Preflight Training - None

3.4 Preflight Activities

TABLE 3.4: PREFLIGHT ACTIVITIES

Preflight Activity				
Description:	Testing at L-9/6 months will include a nasal screen for MRSA. Using a double swab, both nares will be sampled and positive isolates tested for antibiotic susceptibility.			
Schedule:	**Duration:**	**Schedule:**	**Flexibility:**	**Personnel Required:**
	Crew Microbiology - MAT - 15 min	L-9/6 months	Contact lab if accommodation is needed outside existing schedule flexibility	Crewmembers/Lab Personnel
Ground Support Requirements Hardware/Software	**Preflight Hardware:**	**Preflight Software:**	**Test Location:**	
	Crew Sample Collection Kit (double culture swabs with liquid Stuart's Transport Media)	N/A	U.S.	
Testing Facilities	**Minimum Room Dimensions:**	**Number of Electrical Outlets:**	**Temperature Requirements:**	**Special Lighting:**
	8' x 10' (typical exam room)	N/A	Ambient	N/A
	Hot or Cold Running Water:	**Privacy Requirements:**	**Vibration/Acoustic Isolation:**	**Other:**
	Sink with hot and cold water to wash hands	Private room free of distractions	N/A	1 table, 1 chair
Constraints/Special Requirements:	Specimens should be kept between 2°C and 25°C during transport if transport time will be less than 48 hours. If transport time will be more than 48 hours, samples should be kept at 2-8°C during transport. Protect samples against freezing or exposure to excessive heat.			
Launch Delay Requirements:	Microbiology testing will be repeated at flight surgeon's discretion if launch is delayed.			
Notes:	N/A			
Data Delivery	A final report will be transferred to the EMR 24-72 hours following sample receipt.			

3.5 In-Flight Activities - None

3.6 Postflight Activities

TABLE 3.6: POSTFLIGHT ACTIVITIES

Postflight Activity				
Description:	Testing at R+0 d (Perform at R+3 if sample cannot be collected at R+0) will include a nasal screen for MRSA. Using a double swab, both nares will be sampled and positive isolates tested for antibiotic susceptibility. Testing at R+1 d for Direct Returning crewmembers will include a nasal screen for MRSA.			
Schedule:	**Duration:**	**Schedule:**	**Personnel Required:**	
	Crew Microbiology- MAT - 15 min	R+0 d (Collect at R+3 d if sample cannot be collected at R+0) (For direct return, collect in Crew Quarters at R+1)	Crewmembers/Lab personnel	
Ground Support Requirements Hardware/Software	**Postflight Hardware:**	**Postflight Software:**	**Test Location:**	
	Crew Sample Collection Kit (double culture swabs with liquid Stuart's Transport Media)	N/A	Russia/U.S.	
Testing Facilities	**Minimum Room Dimensions:**	**Number of Electrical Outlets:**	**Temperature Requirements:**	**Special Lighting:**
	8' x 10'	2 (U.S. 110V, Russia 220V)	Ambient	N/A
	Hot or Cold Running Water:	**Privacy Requirements:**	**Vibration/Acoustic Isolation:**	**Other:**
	Sink with hot and cold water to wash hands	Private room free of distractions	N/A	1 table, 1 chair
Constraints/Special Requirements:	Post return, testing should occur, as close as possible, to R+0 to reflect ISS conditions, rather than terrestrial conditions. R+1 direct return samples are obtained at JSC Crew Quarters. For collection in Russia, specimens must be returned to JSC Microbiology Laboratory within 48 hours to the greatest extent possible. Specimens should be kept between 2°C and 25°C during transport if transport time will be less than 48 hours. If transport time will be more than 48 hours, samples should be kept at 2-8°C during transport. Protect samples against freezing or exposure to excessive heat.			
Early Destow / Early Return:	N/A			
Notes:	R+0 samples are collected in Russia and returned with other laboratory specimens on crew transport aircraft to Houston. R+1 direct return samples are collected at JSC Crew Quarters and processed in the JSC Clinical Laboratory.			
Data Delivery	A final report will be transferred to the EMR 24-72 hours following sample receipt.			

3.7 Summary Schedule

TABLE 3.7: SUMMARY SCHEDULE

ACTIVITY	DURATION	SCHEDULE	FLEXIBILITY	PERSONNEL REQUIRED	CONSTRAINTS
Preflight Training - None					
Preflight					
Crew Microbiology - MAT	15 min	L-9/6 months	Contact lab if accommodation is needed outside existing schedule flexibility	Crewmember/Lab Personnel	Nasal screening will take place at JSC. Specimens should be kept between 2°C and 25°C.
In-Flight Activity - none					
Postflight -					
ACTIVITY	**DURATION**	**SCHEDULE**	**FLEXIBILITY**	**PERSONNEL REQUIRED**	**CONSTRAINTS**
Crew Microbiology - MAT	15 min	R+0 d (Perform at R+3 d if l sample cannot be collected at R+0) (For direct return, collect in Crew Quarters at R+1)	Contact lab if accommodation is needed to schedule	Crewmember/Lab Personnel	• Post return, testing should occur, as close as possible, to R+0 to reflect ISS conditions. • R+1 direct return samples are obtained at JSC Crew Quarters. For collection in Russia, specimens must be returned to JSC Clinical Laboratory within 48 hours to the greatest extent possible. Specimens should be kept between 2°C and 25°C during transport if transport time is less than 48 hours. If transport time will be more than 48 hours, samples should be kept at 2-8°C during transport. Protect samples against freezing or exposure to excessive heat.
Postflight Debrief – N/A					